DIRECT IMAGE OF A BOUNDED BELOW COHERENT COMPLEX WITH PROPER SUPPORTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-perfect.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the counted statement is the author lemma environment labelled lemma-direct-image-coherent-bdd-below; the proof environment immediately following it is its own proof; the cited sibling lemma is proved in full in the same section (and is a public corpus entry)

Editorial note. The label lemma-direct-image-coherent-bdd-below names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the bounded case lemma of the same section (lemma-direct-image-coherent): for f locally of finite type with Y Noetherian, E in D^b_Coh with H^i(E) proper over Y for all i, Rf_*E is in D^b_Coh, proved by the spectral sequence R^p f_* H^q(E) => R^{p+q} f_*E
- the truncation distinguished triangles tau_{le a} E to E to tau_{ge a+1} E to tau_{le a} E[1] and exactness of Rf_*

Source reference. Original reference 08E2: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 1411–1411. Prerequisite pointer invoked at proof line(s) 1448; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume f is locally of finite type and Y is Noetherian. Let E be an object of $D^+_{Coh}(\mathcal{O}_X)$ such that the support of $H^i(E)$ is proper over S for all i. Then Rf_*E is an object of $D^+_{Coh}(\mathcal{O}_Y)$.*

Proof. The proof is the same as the proof of Lemma 08E2. You can also deduce it from Lemma 08E2 by considering what the exact functor Rf_* does to the distinguished triangles $\tau_{\leq a}E \to E \to \tau_{\geq a+1}E \to \tau_{\leq a}E[1]$. □

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